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Group 5 Project

HydreMine

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1. Initial Proposal

Parts List:

- 2 RC cars
- an Arduino
- 16 Hall effect sensors
- 2 Rumble motors
- 1 Piezo buzzer
- A thin wooden plate
- 2 Magnets

Setup:

An Arduino is connected to roughly 16 Hall effect sensors which are fastened to the bottom of a 2x2m thin, wooden plate. The same Arduino also has a piezo buzzer, and two rumble motors connected. The rumble motors are attached to the bodies of the players. Finally, magnets will be placed on both of the RC cars. One of the RC cars has the south pole of the magnet facing down, while the other has the north pole of the magnet facing down.

Game Description:

Two players drive RC cars around the playing field (wooden plate) looking for “mines” to disarm. However, the mines aren’t visible to the players. Instead, the rumble motors attached to the players will vibrate more intensely as their cars get closer to a mine. When a player finds a mine, a sound will play indicating that the mine has been found, the player who found it gets a point added to their score, and a new mine gets activated for the players to find. This is a competitive game in which the players try to find more mines than each other within a specific time frame. The end of the game is announced by a repeated beeping of the piezo buzzer (akin to a flute whistle at the end of a football match), at which point the player with the highest score wins

2. Who Did What

We mostly worked as a team in this project. Discussing every step with all members involved, especially in the beginning. When we did the initial testing and code setup we all had a clear sense of the end product and we had worked through the initial challenges. Timothy took it upon himself to complete the code in his own time and after we had secured all the necessary components we planned to start implementing them in the final design. Unfortunately, Mats got sick just before the 1st day of code implementation and wiring and could not attend. The majority of the project was done after that day, but at the presentation day the game was not working as expected and the whole team stayed to complete it.

Ideation:	Proposal:	Physical Setup Design:	Initial Testing and Code:	Code Completion:	1 st Day of Code and Wiring:	2 nd Day of Code and Wiring:	Documentation:
Mats Timothy Payman	<u>Payman</u> Timothy Mats	Mats Timothy Payman	Mats Timothy Payman	<u>Timothy</u>	Timothy Payman	Mats Timothy Payman	<u>Mats</u> Timothy Payman

*Underscore means a lead role

3. Problems and Solutions

We initially had an idea involving custom made electromagnets as mines under the playfield and a hall effect sensor on the car to measure them, but we decided not to continue with this approach because we thought it would be too time consuming

We found out that the range of hall effect sensors was quite short. Our first plan was to use a very thin wooden platen, but this would not have worked because of the strength needed to support the cars as well as the weight of the wood itself. We settled on a 9 mm wooden plate, and we ended up drilling holes in it to increase range.

Finding the right RC car was difficult because speed would drastically impact the playability of our game. This was due to the short amount of time the sensors would have to sense the magnets.

Finding the right magnet was also quite difficult. We wanted to have a magnet with a clear north and south pole and a strong magnetic field. This was also due to the realisation that the sensors had less range than anticipated. Eventually we used neodymium, disc-shaped magnets. In replications of this project even bigger magnets can be used to generate a larger effect on the sensors.

When building the playing field, we discovered that we did not have enough long wires. We were able to borrow some copper tape from a classmate. Using this tape, we were able to make up for the short wire length by connecting the wires to the tape.

The rumble motors were finnickier to work with and often stopped working due to wiring issues. By soldering the wires to the rumble motors, we were able to extend the time that the rumble motors worked. Nonetheless, they broke relatively soon regardless.

The reading of the hall effect sensors was inconsistent at times, leading to the rumble motors not activating properly. By changing the code so that it would set an average baseline for each sensor, the consistency improved greatly. This was an important realization because for a short while we thought our project might be doomed.

The size of the playing field was essential because of the speed of the cars, but this made it hard to work on the wiring, as it was on the bottom side of the playing field which meant that we had to pick it up and place it against a wall anytime we wanted to test it. We placed the playing field on some chairs so that we would be able to slide under the playing field and make changes like that.

We initially planned to use 16 hall effect sensors. To accomplish this with a single Arduino we had to work with a multiplexor. During testing we broke two of these multiplexors. After this we had the idea to use multiple Arduinos and have them communicate, but after researching how to do this we decided that it would be too complex and moved on with one Arduino.

4. Result

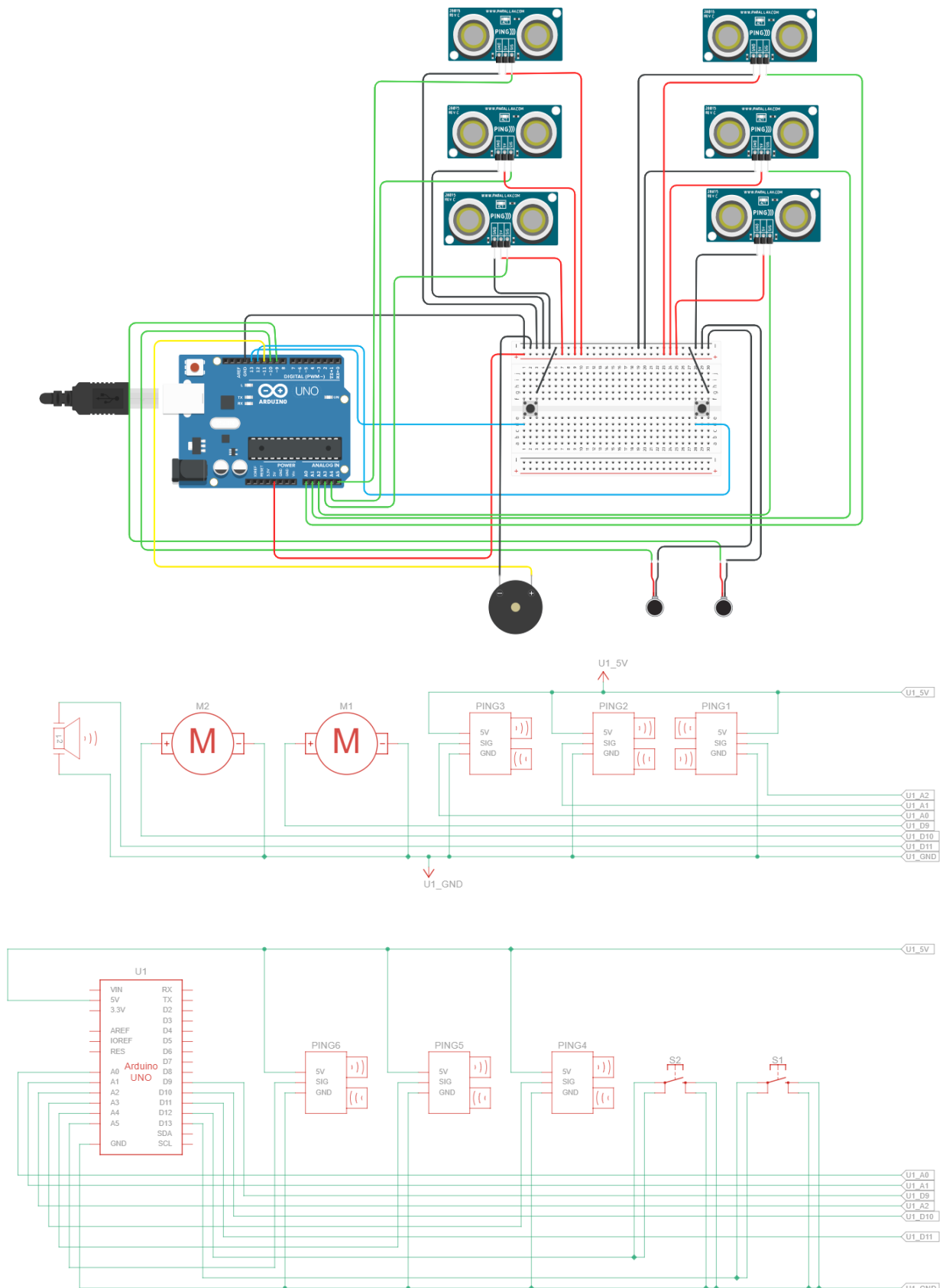
The result is a fully working game where two players use RC cars to find mines on a 2 by 2 meter playing field. Each player is hooked up to a vibration motor, and has a button placed before them. A magnet is placed on both cars, one car has the north-pole facing down, the other car has the south-pole facing down. This difference in polarity is picked up by 6 different hall effect sensors placed under the playing field (mines). A random seeds dictates which mine is active. Both players try to find the active mine before the other. They do this by hovering over the 6 different holes. If a player is hovering over the active mine with their car, they will feel a vibration of the motor. If this player then pushes his or her button they will earn a point. Feedback on earned points is provided through a buzz from a piezo. The first player to get three points is the winner. The piezo will then give three buzzes to resemble a whistle and the game starts again.

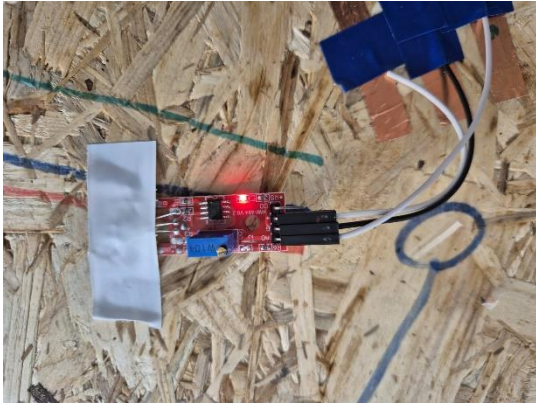
5. Evaluation

We had to work through a number of difficulties to get to the final product. Most of the things we changed were after realizing the limitations of the hall effect sensors. We wanted to make the mines invisible for the players, and make it so they could only be located by the vibrations of the vibration motors, but because of the short speed of the cars and the range of the sensors this was not possible. We also had to change the size and thickness of the playing field, as well as drill holes in it to improve the range of the sensors, further revealing the location of the mines. We thought this would influence the enjoyment of the game, but this was not the case. It proved quite challenging and fun to find the mines before your opponent. Another big change was the transition from 16 to 6 sensors. This was due to hardware problems with the multiplexor. This proved to be a good thing because 16 mines would have been way too many to cycle through before the active mine would be found.

6. Figures and Images

In the first two figures it should be noted that the hall effect sensors are replaced with other sensors. This is because the sensors could not be displayed in the program that was used. The wiring is identical with the use of hall effect sensors.

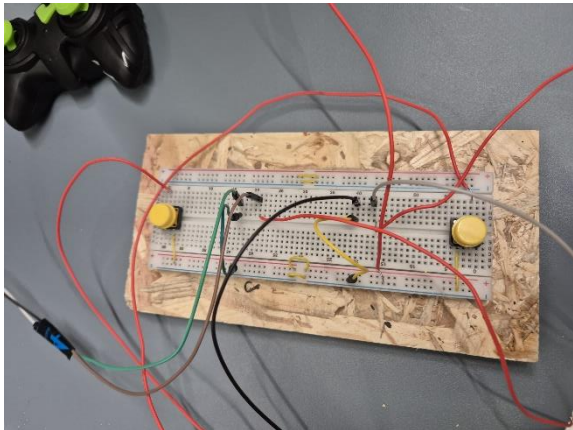




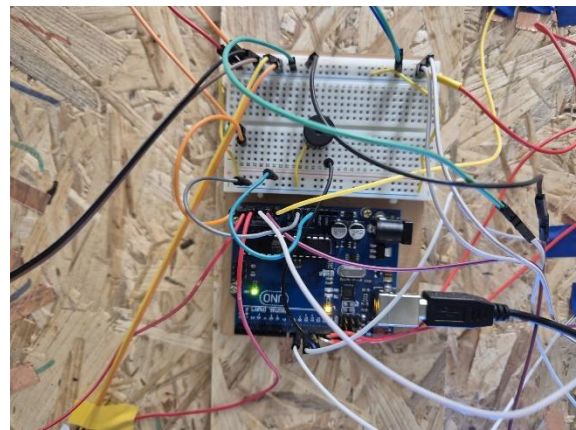
Hall effect sensor strapped under one of the hole



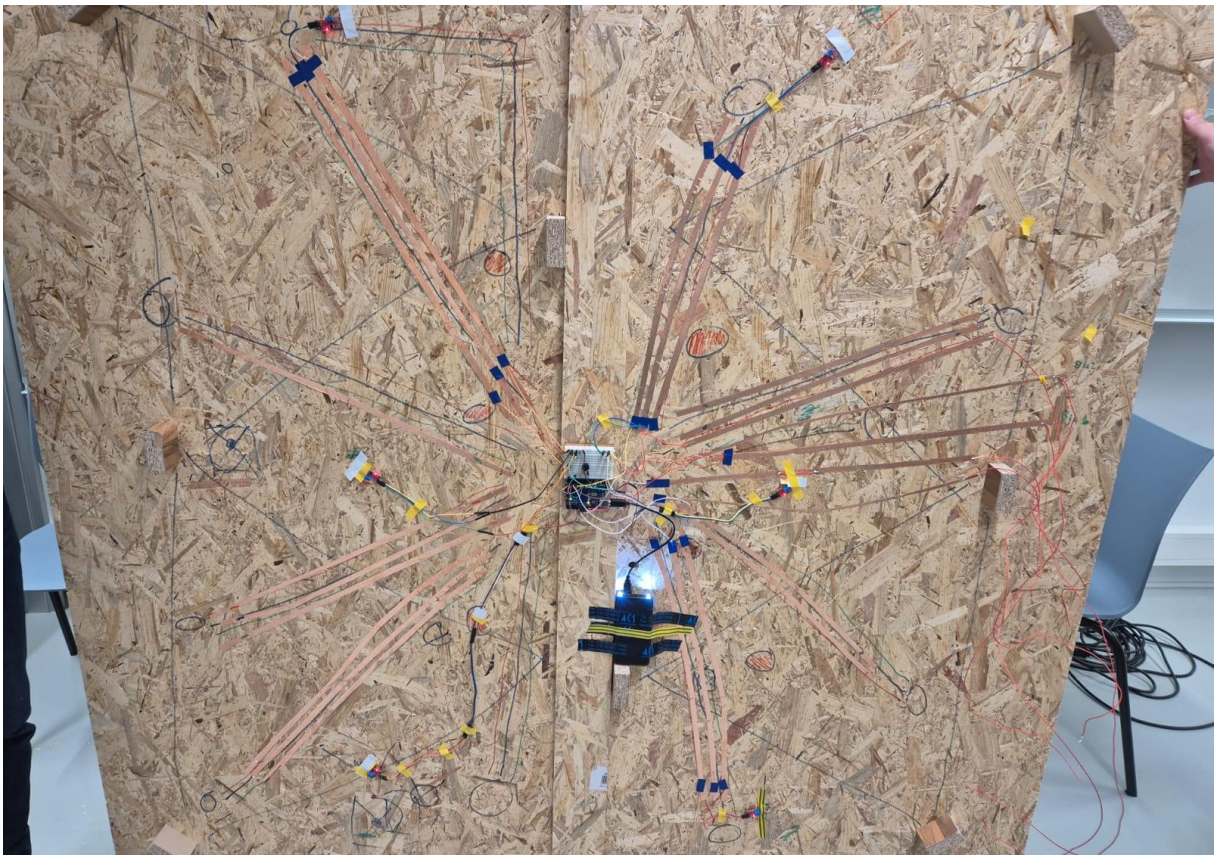
Arduino was running on a powerbank without a laptop present



Close up of the buttons and vibration motor connections



Close up of the Arduino, all incoming wires and the piezo buzzer



Overview of the wiring underneath the whole playing field